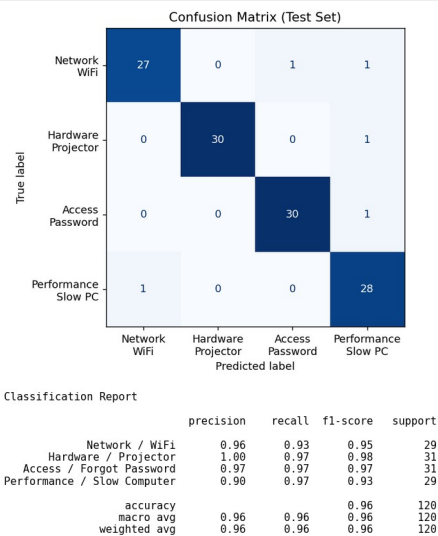
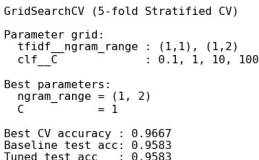


Project Development Phase Model Performance Test

Date	02 October 2026
Team ID	SWTID-2026-5730
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot																																																																	
1.	Metrics	Classification Model (Multinomial text classification – TF-IDF + Logistic Regression): Dataset: 600 tickets (Train 480 / Test 120); 4 classes – Network/WiFi, Hardware/Projector, Access/Forgot Password, Performance/Slow Computer. Confusion Matrix - rows = actual, columns = predicted (order: WiFi, Projector, Password, Slow PC) <pre>[27 0 1 1] [0 30 0 1] [0 0 30 1] [1 0 0 28]</pre> Accuracy Score - 95.83% (0.9583) Classification Report - Macro avg Precision 0.96, Recall 0.96, F1-score 0.96; per-class F1: WiFi 0.95, Projector 0.98, Password 0.97, Slow PC 0.93. Regression Model: Not applicable (problem is classification).	 The screenshot displays the Confusion Matrix (Test Set) and the Classification Report for the Classification Model. The Confusion Matrix shows the relationship between True labels (Network WiFi, Hardware Projector, Access Password, Performance Slow PC) and Predicted labels. The Classification Report provides performance metrics (precision, recall, f1-score, support) for each class and overall averages (accuracy, macro avg, weighted avg). <table border="1"><caption>Confusion Matrix (Test Set)</caption><thead><tr><th>True label \ Predicted label</th><th>Network WiFi</th><th>Hardware Projector</th><th>Access Password</th><th>Performance Slow PC</th></tr></thead><tbody><tr><th>Network WiFi</th><td>27</td><td>0</td><td>1</td><td>1</td></tr><tr><th>Hardware Projector</th><td>0</td><td>30</td><td>0</td><td>1</td></tr><tr><th>Access Password</th><td>0</td><td>0</td><td>30</td><td>1</td></tr><tr><th>Performance Slow PC</th><td>1</td><td>0</td><td>0</td><td>28</td></tr></tbody></table> <table border="1"><caption>Classification Report</caption><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>Network / WiFi</td><td>0.96</td><td>0.93</td><td>0.95</td><td>29</td></tr><tr><td>Hardware / Projector</td><td>1.00</td><td>0.97</td><td>0.98</td><td>31</td></tr><tr><td>Access / Forgot Password</td><td>0.97</td><td>0.97</td><td>0.97</td><td>31</td></tr><tr><td>Performance / Slow Computer</td><td>0.90</td><td>0.97</td><td>0.93</td><td>29</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.96</td><td>120</td></tr><tr><td>macro avg</td><td>0.96</td><td>0.96</td><td>0.96</td><td>120</td></tr><tr><td>weighted avg</td><td>0.96</td><td>0.96</td><td>0.96</td><td>120</td></tr></tbody></table>	True label \ Predicted label	Network WiFi	Hardware Projector	Access Password	Performance Slow PC	Network WiFi	27	0	1	1	Hardware Projector	0	30	0	1	Access Password	0	0	30	1	Performance Slow PC	1	0	0	28		precision	recall	f1-score	support	Network / WiFi	0.96	0.93	0.95	29	Hardware / Projector	1.00	0.97	0.98	31	Access / Forgot Password	0.97	0.97	0.97	31	Performance / Slow Computer	0.90	0.97	0.93	29	accuracy			0.96	120	macro avg	0.96	0.96	0.96	120	weighted avg	0.96	0.96	0.96	120
True label \ Predicted label	Network WiFi	Hardware Projector	Access Password	Performance Slow PC																																																																
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2.	Tune the Model	Hyperparameter Tuning - GridSearchCV over TF-IDF ngram_range {(1,1),(1,2)} and Logistic Regression C {0.1, 1, 10, 100}. Best: ngram_range=(1,2), C=1. Validation Method - 80/20 stratified hold-out split plus 5-fold Stratified Cross-Validation on the training set (best CV accuracy 96.67%). Baseline test accuracy 95.83% vs tuned 95.83%.	 The screenshot shows the results of GridSearchCV (5-fold Stratified CV). It lists the parameter grid (tfidf_ngram_range: (1,1), (1,2) and clf_C: 0.1, 1, 10, 100), the best parameters (ngram_range = (1, 2) and C = 1), and the best CV accuracy (0.9667), baseline test accuracy (0.9583), and tuned test accuracy (0.9583).																																																																	